

Seungyeop Lee

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RESEARCH INTERESTS

Lower-limb wearable robot Vision-based perception Human-robot interaction

EDUCATION

Korea University

M.S. in Mechanical Engineering. Integrated Major in Intelligent Mobility.

Research Lab: Human Machine Systems Lab (Prof. Shinsuk Park)

Thesis: A Development of Intuitive Controller and Monocular Vision based Interface for Disaster Response UAV

Seoul, Republic of Korea

Sep. 2022 – Feb. 2025

Korea University

B.S. in Mechanical Engineering.

(Note: Absence during Oct. 2017 – Jun. 2019 due to military service)

Seoul, Republic of Korea

Mar. 2016 – Aug. 2022

RESEARCH EXPERIENCE

Robot Vacuum Cleaner SW Lab – Perception Team, Samsung Electronics

Software Engineer, Spatial AI Development

Suwon, South Korea

May 2026 – Present

Spatial AI for Resource-Constrained Robot Vacuum Cleaners

- Developed a dual-process robot intelligence architecture inspired by human cognition combining task knowledge reuse and incremental scene-graph enrichment to reduce repeated reasoning on resource-constrained robots.
- Developed a viewpoint selection algorithm that selects navigation coordinates for robots to acquire object information based on visibility and accessibility.
- Developed room classification using object occurrence likelihoods and spatially accumulated visual evidence from observations collected during navigation.
- Integrated VLM, perception, and navigation into a unified Spatial AI system and validated the system in NVIDIA Isaac Sim.

Wearable Robotics Laboratory

Research Intern, Advisor: Prof. Jinsoo Kim

Seoul National University, Seoul, South Korea

Dec. 2024 – Feb. 2025

Embedded Controller and Sensor Integration for a Soft Exosuit

- Developed a real-time onboard controller for a tendon-driven soft exosuit.
- Implemented a 10-ms sensor-to-motor control loop for a tendon-driven soft exosuit using multithreaded CAN communication.
- Designed a sensor integration module connecting wearable sensors, actuators, and the controller.

Human-Machine Systems Lab

Graduate Researcher, Advisor: Prof. Shinsuk Park

Korea University, Seoul, South Korea

Sep. 2022 – Dec. 2024

Monocular Vision-Based Assistance Interface for Indoor Drones

- Developed a monocular vision-based pilot assistance interface to reduce human error during indoor UAV operation.
- Applied monocular depth estimation for obstacle perception using a single RGB camera on a lightweight UAV.
- Evaluated the interface with participants: NASA-TLX mental demand decreased by 43%, and mean System Usability Scale scores increased from 62.5 to 80 versus the baseline interface.

Intelligent Machine Perception and Learning (IMAPLE) Lab

Visiting Researcher, Advisor: Prof. David Han

Drexel University, Philadelphia, PA, USA

Sep. 2023 – Feb. 2024

Sim-to-Real Monocular Depth Estimation

- Applied CycleGAN-based domain transfer to generate realistic training data from synthetic images.
- Improved monocular depth estimation using mixed real and domain-transferred synthetic data.

Autonomous Lighter-Than-Air Robot

- Developed an autonomous aerial robot using STM32 PID control and YOLOv5-based perception.

INDUSTRY EXPERIENCE

Samsung Electronics, Robot Vacuum Cleaner Development Lab

Software Engineer, Robot Software Development

Suwon, South Korea

May 2026 – Present

Hardware-in-the-Loop Simulation and Validation

- Built an Isaac Sim-based HIL test system with Samsung R&D Institute-Delhi, integrating a physical controller board and production control software with simulated robot sensors.

Samsung Electronics, Robot Vacuum Cleaner Development Lab

Mechanical Engineer, Robot Product Development

Suwon, South Korea

Feb. 2025 – Apr. 2026

Robot Vacuum Cleaner Mechanism Development and Mass Production

- Designed and validated mop actuation mechanisms for robot vacuum cleaners, including durability improvement.
- Supported mass production and quality validation, including performance testing and vision-based inspection.

PUBLICATIONS & PRESENTATIONS

Seungyeop Lee*, Yeeun Kim*, Sooho Park, Seong Oh Lee, and Jong Jin Park. “**DART: From Attentive Reasoning to Intuitive Execution for Experience-Driven Robot Intelligence.**” Submitted to the *2027 IEEE International Conference on Robotics and Automation (ICRA)*. Under review.

Knut Peterson*, **Seungyeop Lee***, Solmaz Arezoomandan, and David Han. “**Domain-Transferred Synthetic Data Generation for Improving Monocular Depth Estimation**”, *2025 25th International Conference on Control, Automation and Systems (ICCAS)*, 2025.

Seungyeop Lee, and Shinsuk Park. “Obstacle Avoidance for Small Unmanned Aerial Vehicles Using Deep Learning-Based Monocular Depth Estimation”, *Institute of Control, Robotics and Systems Conference (ICROS)*, 2024. Poster presentation.

Seungyeop Lee, and Shinsuk Park. “Development of a ROS-Based Indoor Autonomous Quadcopter Platform Using Visual Odometry”, *Institute of Control, Robotics and Systems Conference (ICROS)*, 2023. Poster presentation.

SELECTED PROJECTS & LEADERSHIP

Horang-Nabi Model Aircraft Club Foundation <i>Founder & President</i>	Korea University, South Korea Aug. 2019 – Dec. 2021
• Founded Korea University's first model aircraft club and led UAV projects, operations, and undergraduate study activities.	
Korea Robot Aircraft Competition <i>Computer Vision Development</i>	Korea Aerospace Industries Association, South Korea Apr. 2021 – Sep. 2021
• Developed a pedestrian-tracking UAV for a safe-walk companion service, combining autonomous user tracking with onboard perception and flight control.	
National University Student Model Aircraft Competition <i>Team Leader</i>	Gyeongsang National University, South Korea Nov. 2019 – Nov. 2020
• Led a team to design and build a scale aircraft for the 21st national university model aircraft competition.	
Social Economy Engineering Ideas Competition <i>Mechanical Design</i>	University of Seoul, South Korea Sep. 2020 – Oct. 2020
• Proposed a foldable stair-climbing mobility aid to improve subway accessibility for elderly and mobility-limited users.	
Pi village Start-up Team - DRONEDU <i>Founder</i>	Korea University, South Korea Jan. 2020 – Oct. 2020
• Developed a drone education program for high school and university students as a campus start-up initiative.	

AWARDS AND HONORS

Intelligent Robotics Overseas Research Fellowship <i>Received US\$20,000 in funding from the South Korean government through the Ministry of Trade, Industry and Energy (MOTIE)</i>	Sep. 2023 – Feb. 2024
Future Automobile Industry-university assignment Project Competition <i>Excellence Award, Inter-university Alliance for Future Vehicular Technology (I4FT), South Korea</i>	Jan. 2023
National University Student Model Aircraft Competition <i>Third Prize, Gyeongsang National University, South Korea</i>	Nov. 2020
Social Economy Engineering Idea Competition <i>Third Prize, University of Seoul, South Korea</i>	Oct. 2020
Creative Comprehensive Design Competition <i>Third Prize, Innovation center for engineering education, Korea University, South Korea</i>	Sep. 2020
22nd Pi village Start-up Team - DRONEDU <i>First prize, Korea University, South Korea</i>	Jun. 2020

SKILLS AND TECHNIQUES

Programming Python, ROS1&2, Microprocessor(Jetson, Raspberry Pi, Arduino), Simulation(Isaac Sim, Gazebo)

Hardware 3D CAD(NX, Fusion 360), Rapid prototyping(Laser cutting, 3D printing)

TEACHING AND MENTORING

Teaching Assistant, Dynamics – Korea University • Taught two-link manipulator modeling using Adams simulation.	Sep. 2024 – Nov. 2024
Teaching Assistant, Robotics – Korea University • Taught Adams simulation, Raspberry Pi, and PID-based DC motor control.	Mar. 2023 – Jun. 2023; Mar. 2024 – Jun. 2024
Robotics Career Mentor, Sector Council (SC) Mentoring Program • Led a session in a government-supported career mentoring event as a Samsung Electronics engineer.	Mar. 2026